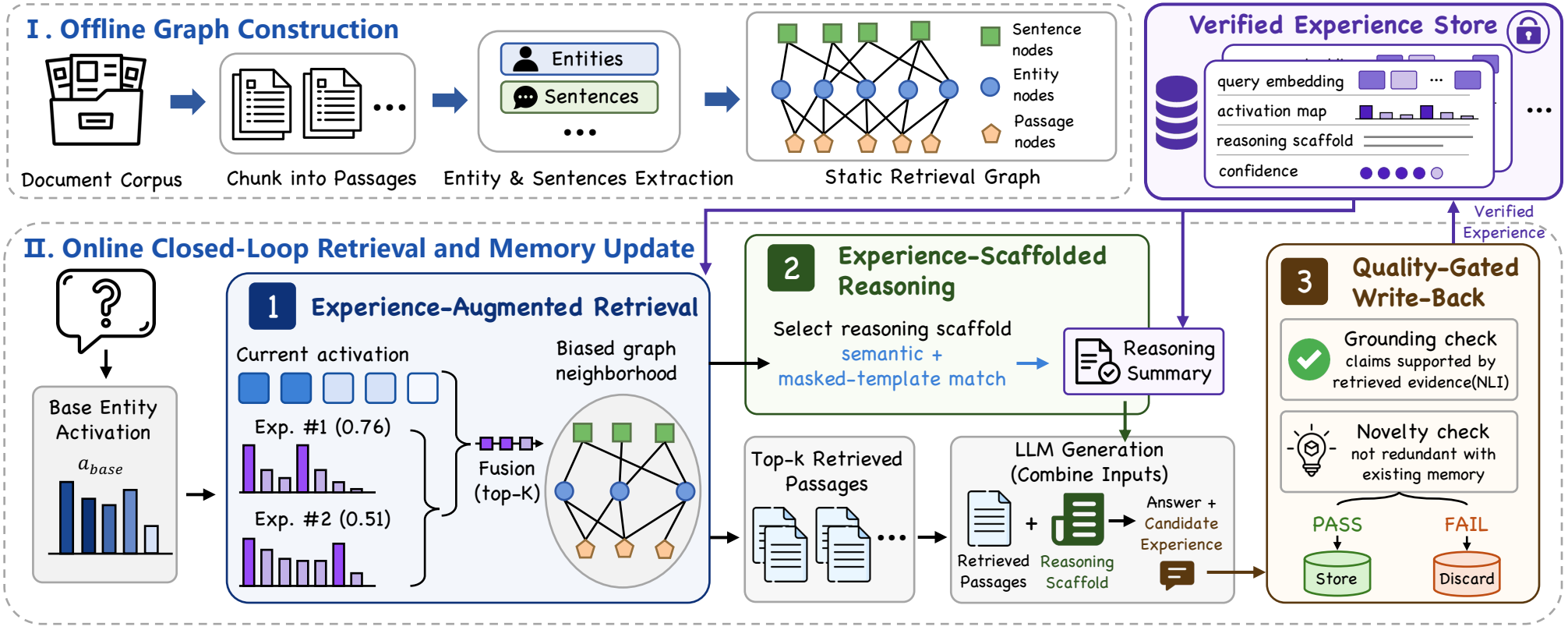

I. Offline Graph Construction
Document Corpus
Chunk into Passages
Entities
Sentences
Entity & Sentences Extraction
Sentence nodes
Entity nodes
Passage nodes
Static Retrieval Graph
Verified Experience Store
query embedding
activation map
reasoning scaffold
confidence
Verified Experience
II. Online Closed-Loop Retrieval and Memory Update
Base Entity Activation
a_{base}
1 Experience-Augmented Retrieval
Current activation
Exp. #1 (0.76)
Exp. #2 (0.51)
Fusion (top-K)
Biased graph neighborhood
2 Experience-Scaffolded Reasoning
Select reasoning scaffold
semantic + masked-template match
Reasoning Summary
Top-k Retrieved Passages
LLM Generation (Combine Inputs)
Retrieved Passages
Reasoning Scaffold
Answer + Candidate Experience
3 Quality-Gated Write-Back
Grounding check
claims supported by retrieved evidence(NLI)
Novelty check
not redundant with existing memory
PASS
FAIL
Store
Discard